

41891: Cloud Computing Infrastructure
42891: Infrastructure for Cloud Computing

Week 5: Data Centre Fundamentals and Virtualisations

Outline

- * What is a Data Center?
 - * Data center goals
 - * Data center facilities
- * Data Center Architecture
- * Virtualising a Data Center
 - * Virtualisation at different layers
 - * Technologies in DC

Introduction to Data Center

- * What is a Data Center?
 - * Data Centers house critical computing resources in controlled environments and under centralized management
 - * These computing resources include mainframes; web and application servers; file and print servers; messaging servers; application software and the operating systems that run them; storage subsystems; and the network infrastructure, etc.

Data Center Goals

- * Traditional business-oriented goals generate a number of IT initiatives:
 - * Business continuance
 - * Increased security in the Data Center
 - * Storage consolidation
 - * Application, server, and Data Center consolidation
 - * Etc.

Data Center Goals

- * Data center design criteria:
 - * Availability
 - * Scalability
 - * Security
 - * Performance
 - * Manageability
 - * Etc.

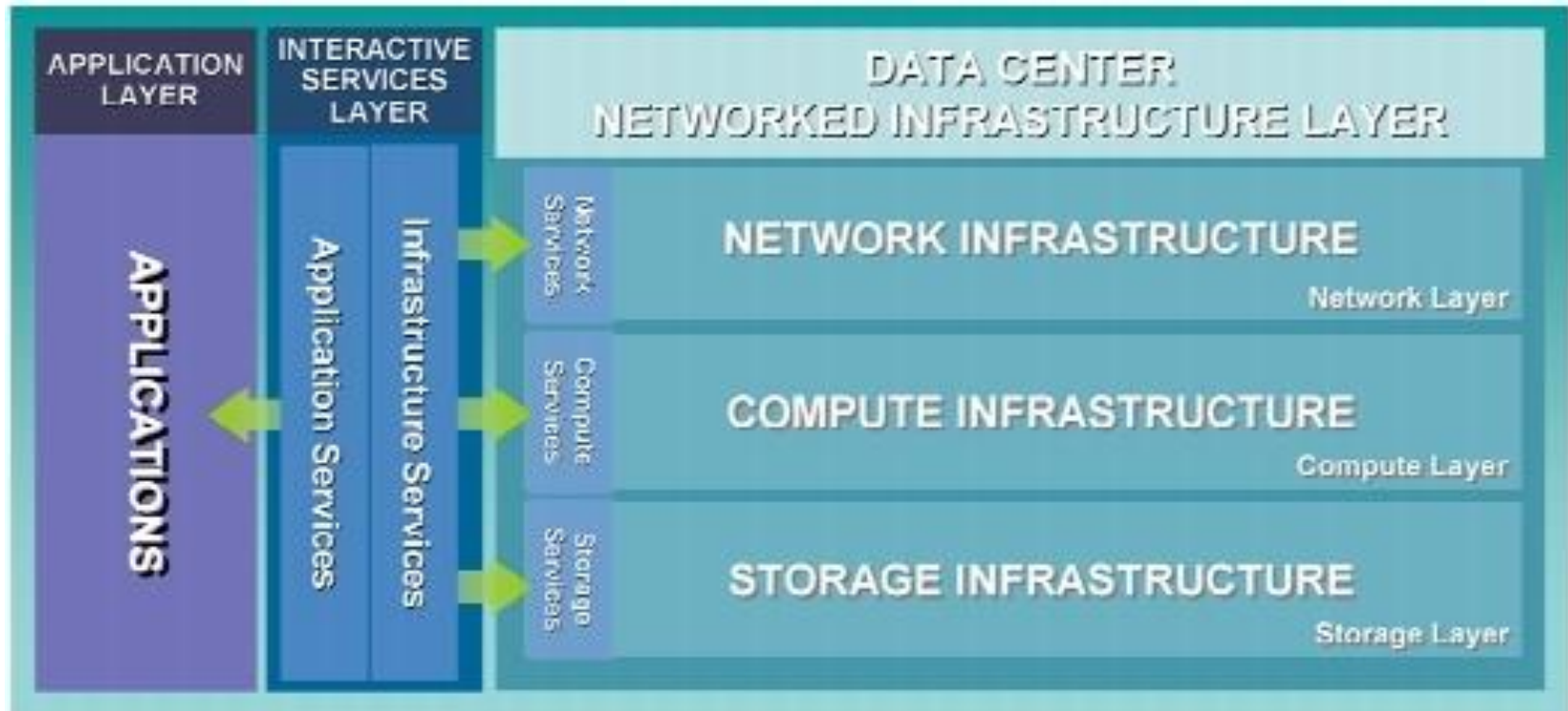
Data Center Facilities

- * There are several considerations to ensure the performance.
 - * Power capacity
 - * Cooling capacity
 - * Cabling
 - * Temperature and humidity controls
 - * Fire and smoke systems
 - * Physical security: restricted access and surveillance systems
 - * Rack space and raised floors
 - * Modularity and flexibility

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- * Virtualising a Data Center

Three-Layer Data Center Model



Why Virtualise?

Is it to save on hardware costs?

Is it because of improved speed of deployment?

Is it because it allows workload mobilisation?

- * Well yes, originally it was?
- * Originally virtualisation was seen as a way run multiple workloads on a single host.

- * Later it was found that it was easier to clone an existing machine than build it from scratch
- * This also meant production machine could be cloned to produce disaster recovery, development and multiple Test environments.

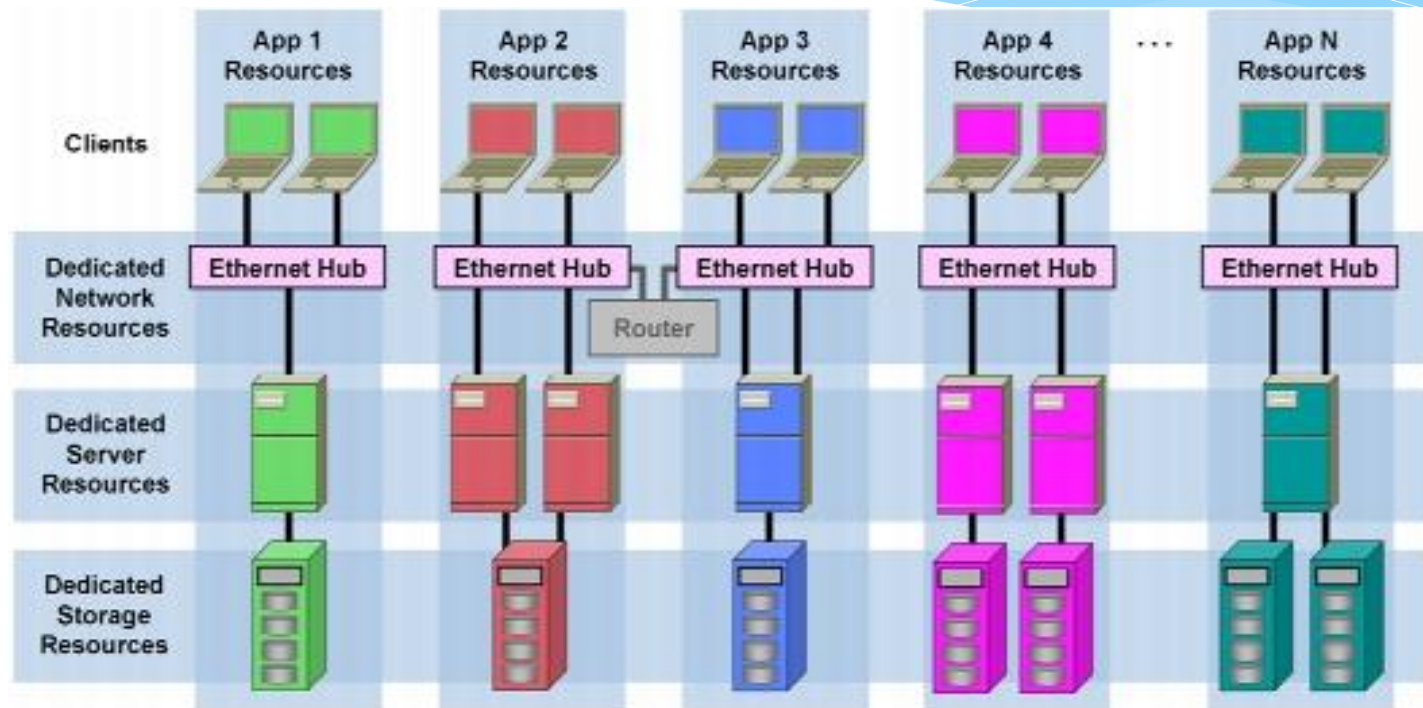
- * This is probably today's most compelling reason.
- * Without workload mobilisation there is no "Cloud"

Benefits of Virtualisation in Data Center

- * Reduced cost
- * Faster redeploy
- * Better testing
- * No vendor lock-in
- * Better disaster recovery
- * Less heat buildup
- * Easier migration to cloud
- * Etc.

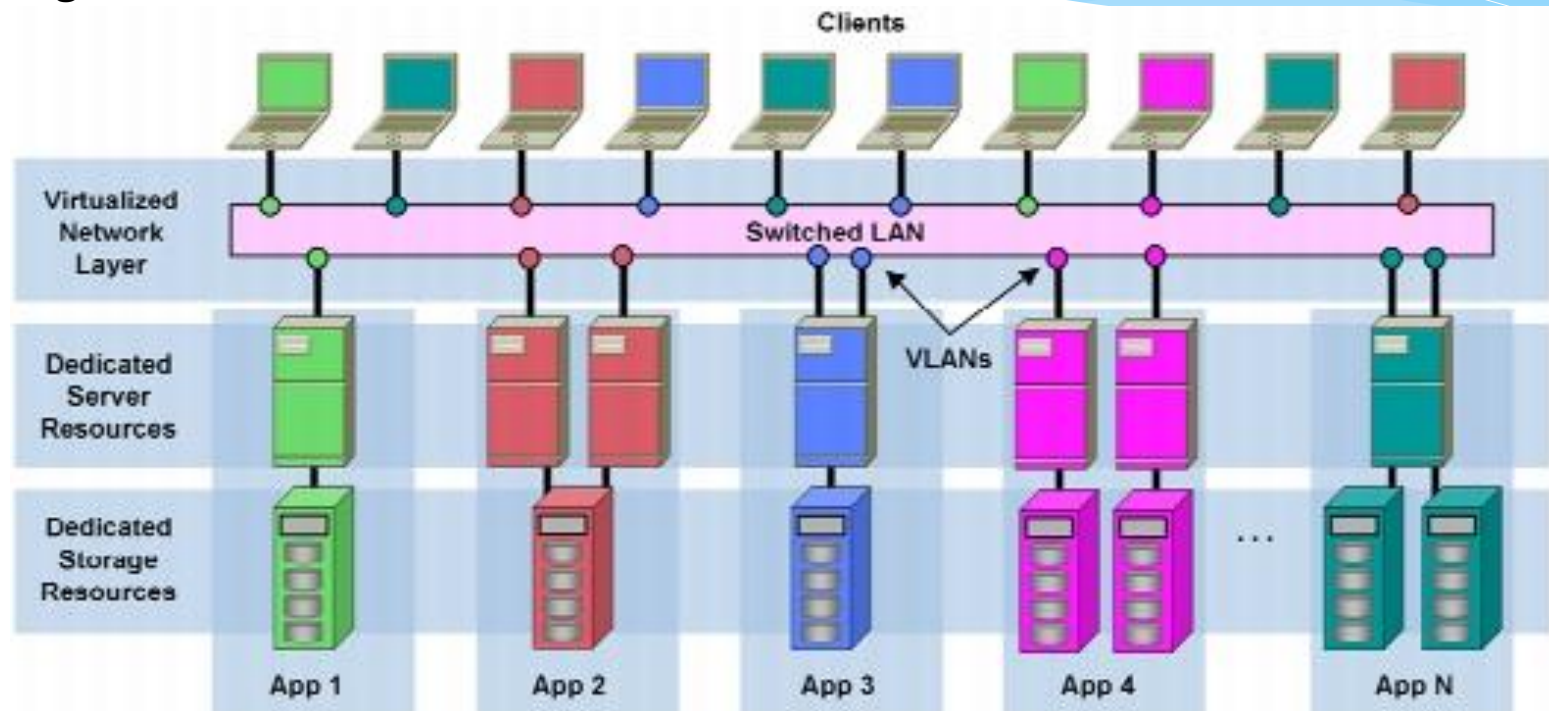
Data Center Architecture with Dedicated Resources

- Dedicated resources at each layer



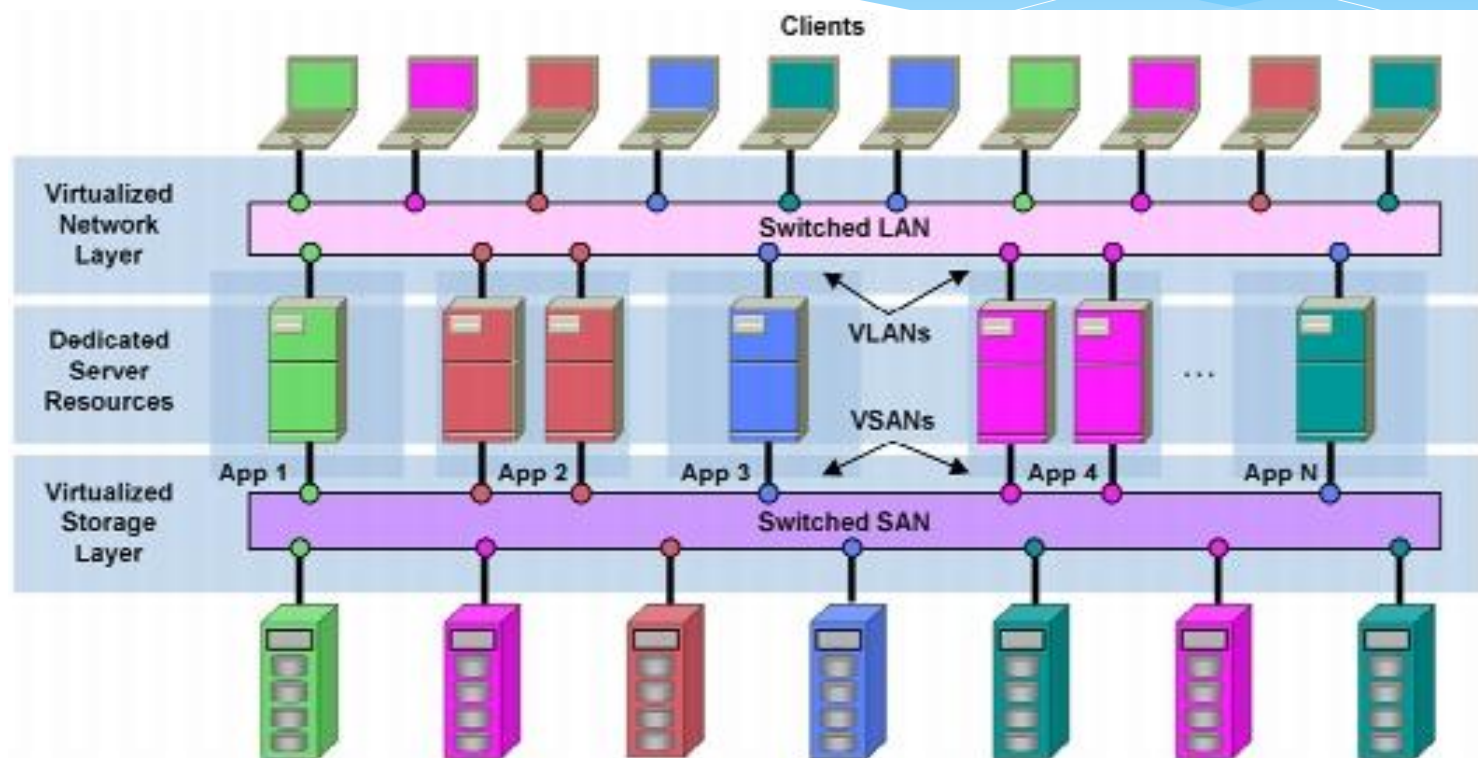
Network Layer Consolidation and Virtualisation

- Network layer consolidation interconnects networking resources into a shared, intelligent network



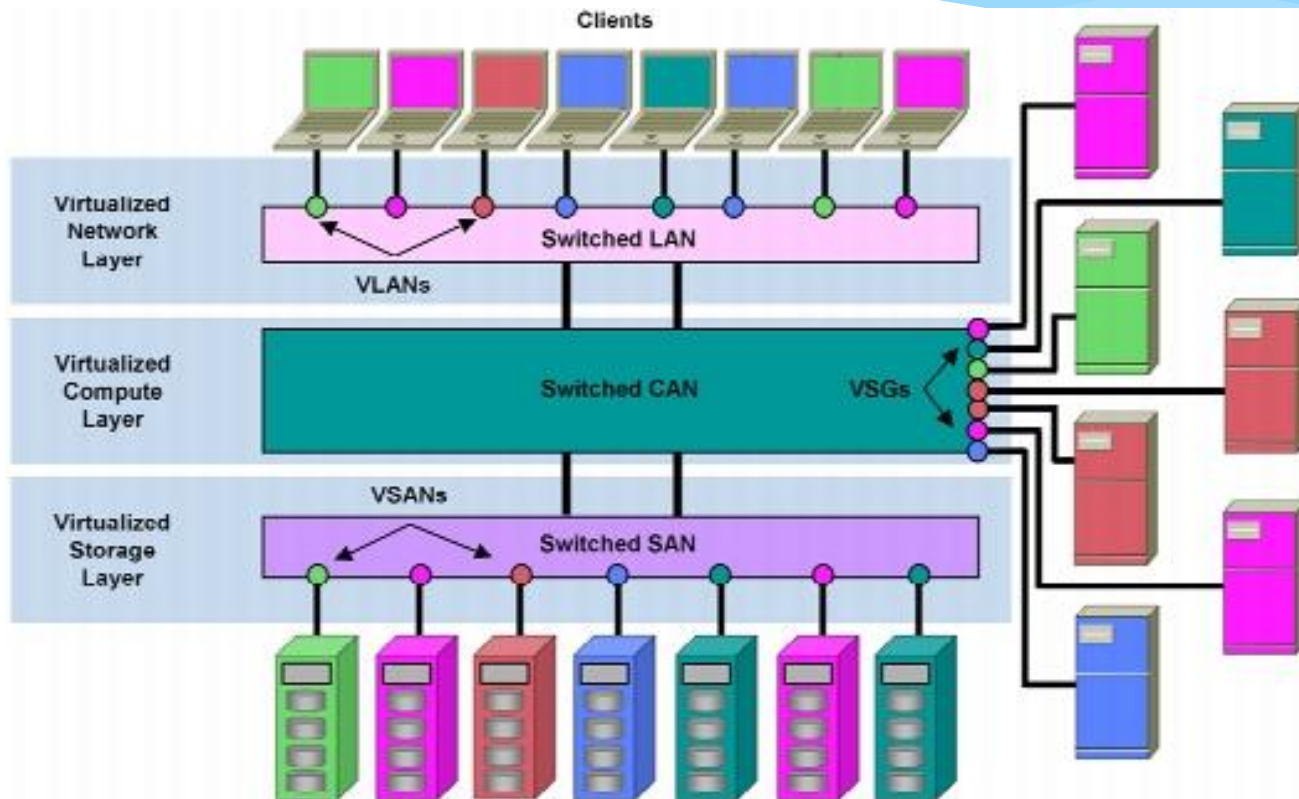
Storage Layer Consolidation and Virtualisation

- Once storage layer is virtualized, storage can be treated as a pooled resource.

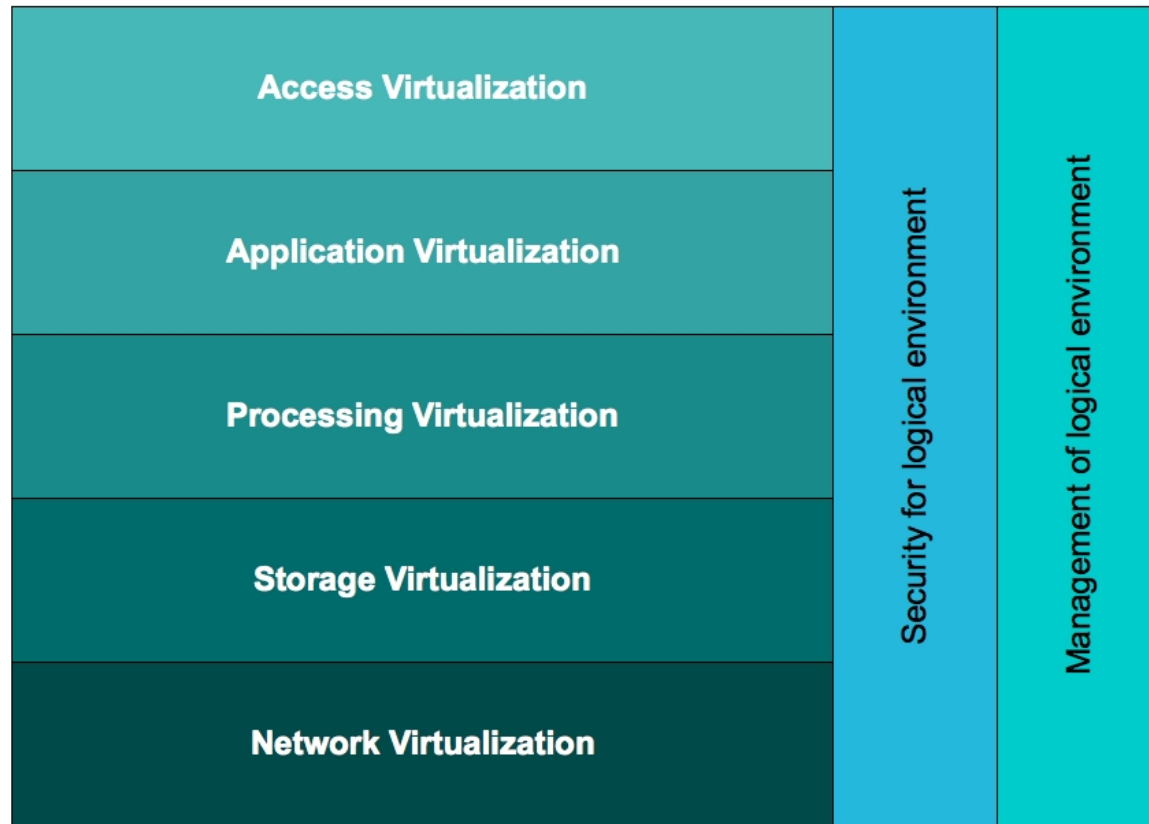


Compute Layer Consolidation and virtualisation

- Compute consolidation reduces costs and increases efficiency of physical server usage by allowing the use of standard servers in pooled configurations.



Kusnetzky Group Model of Virtualisation



Network Virtualisation

- * The process of combining hardware and software network resources and network functionality into a single software-based management entity.
- * Network Virtualisation allows for the aggregation or segregation of logical networks while still utilising one or many physical connections.
- * Common Standards – VXLAN, NVGRE, 802.1q (VLANs)

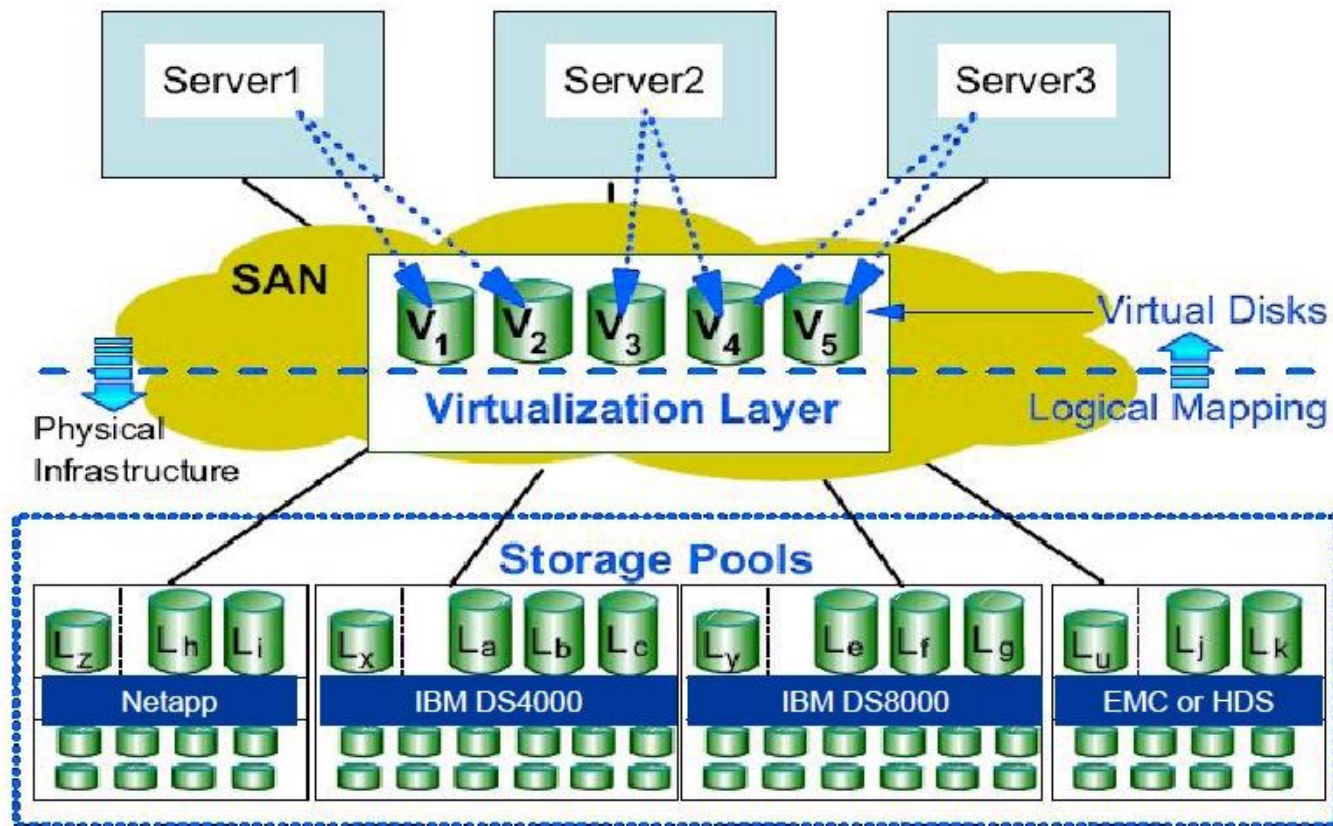
Storage Virtualisation

- * What is storage virtualisation?
 - * A combination of hardware and software technologies that abstracts the physical components of a storage system from the application or server and present it in a consistent manner to the client.
- * Storage Virtualisation is commonly used in virtualised environments to share physical storage systems between multiple virtual machines and hypervisors.
- * Typically Storage Virtualisation starts with a storage device (such as a SAN) that exports the storage either at the Block Level or File Level.
 - * Block level virtualisation: Abstraction of logical storage below the OS or file-system
 - * File level virtualisation: Virtualisation at the file-system level, typically using a network file system.

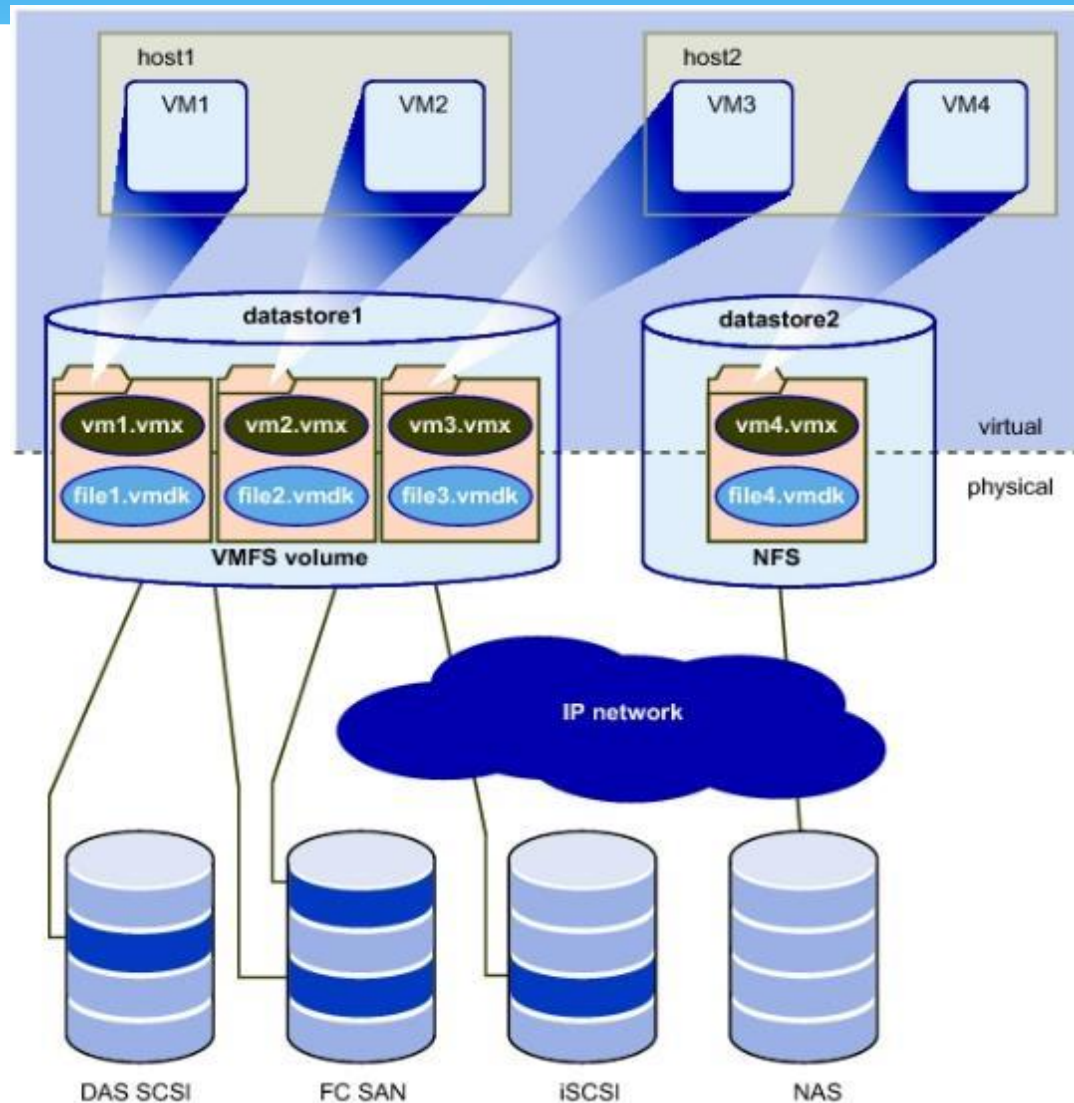
Storage Virtualisation

The function of Storage Virtualisation:

- Allows for multiple systems to access the same storage system.
- Allows for the storage to be physically separated from the local server, but appear to be accessible storage.



VMware Infrastructure Storage Architecture



What is SAN?

- * Storage Area Network (SAN)
 - * A SAN is typically a storage device accessed over a storage network. It is a device that allows you to virtualise multiple physical storage devices into a number of logical storage volumes, often referred to as LUNs.
- * The SAN system keeps track of the mappings between logical and physical storage blocks and provides a translation layer. It allows a general-purpose server to access data on a range of storage devices.
- * Accessed over Fibre Channel, iSCSI, FCoE, ATAoE, etc.

Processing Virtualisation

- * The hardware and software technology that abstracts the physical hardware from system services, operating systems and applications.
- * This type of Virtualization technology can make one system appear to be many or many systems appear to be a single computing resource to achieve goals ranging from raw performance, high levels of scalability, reliability/availability, agility or consolidation of multiple environments onto a single system.
- * Can be used to chop up one server, or join together many to make a grid computer.

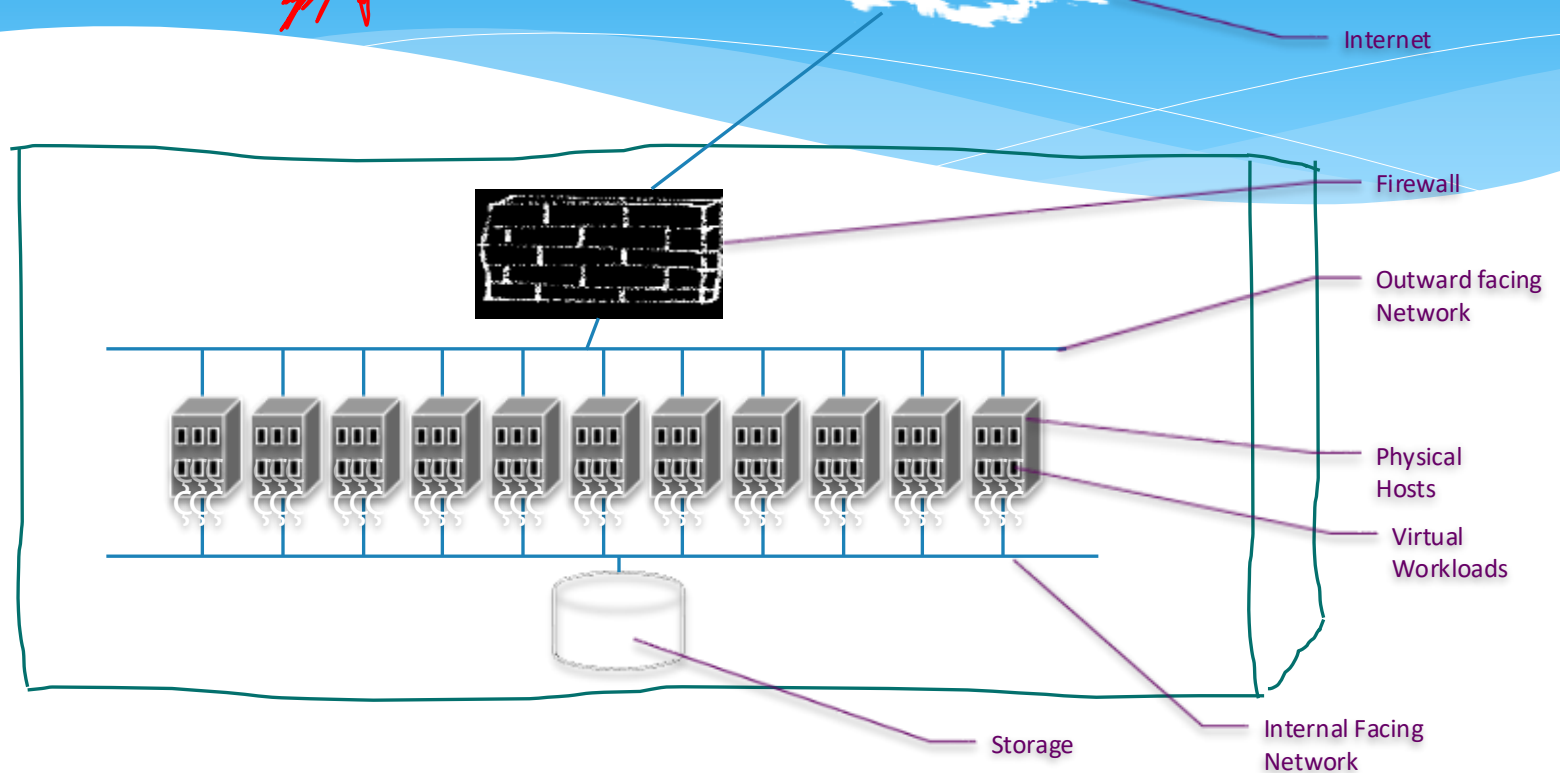
Application Virtualisation

- * Application level virtualisation that allows a piece of software to run on a variety of operating systems, hardware platforms and devices.
- * This usually means that the application has been written to use an application framework.
- * More advanced forms of this technology offer the ability to restart an application in case of a failure, start another instance of an application if the application is not meeting service level objectives, or provide workload balancing among multiple instances of an application to achieve high levels of scalability.
- * E.g. Writing software using the Java Framework allows portability of the code base.

Access Virtualisation

- * Hardware and software technology that allows nearly any device to access any application without either having to know too much about the other.
- * The application sees a device it's used to working with and the device sees an application it knows how to display.
- * In some cases, special purpose hardware is used on each side of the network connection to increase performance, allow many users to share a single client system or allow a single individual to see multiple displays.
- * E.g. Virtual Desktop Infrastructure – utilising protocols such as Microsoft RDP or Citrix ICA

Ideal The Virtual Data Center



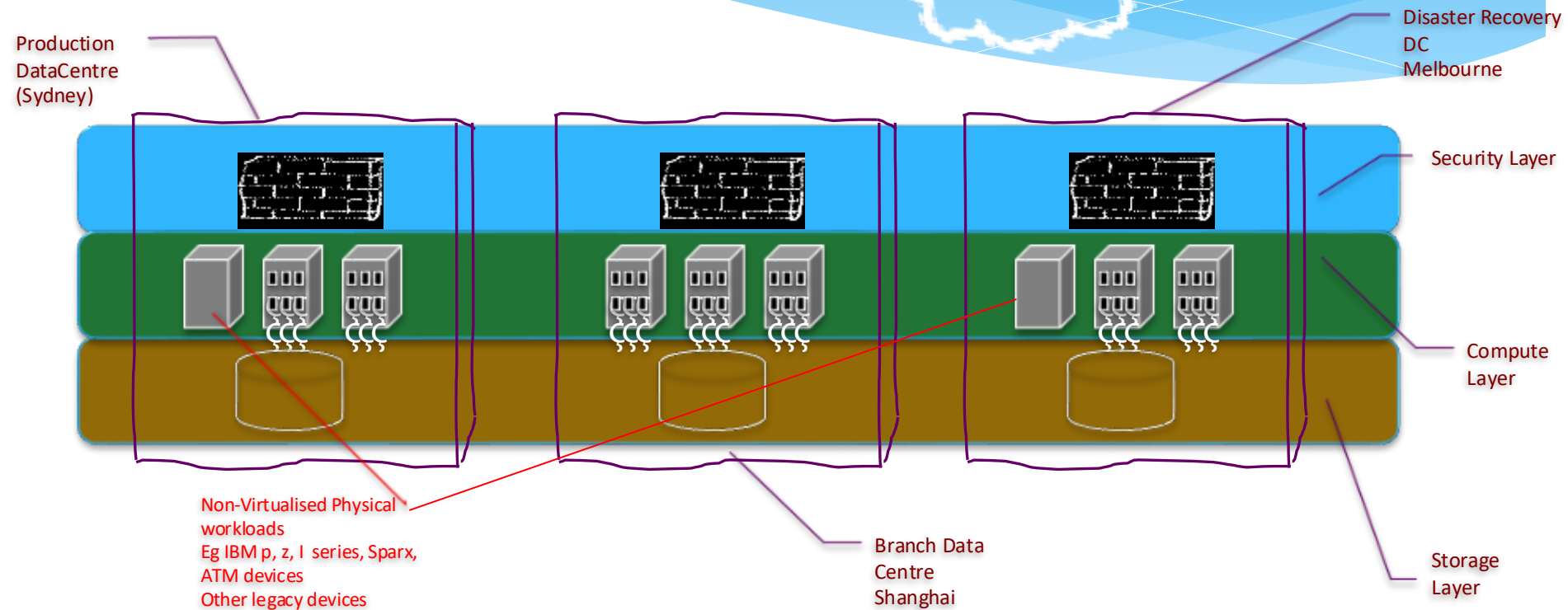


**Is it really that
easy?**



**Umm....
No**

The Virtual Data Center



Technologies in a Data Center

- * Separate Compute, Storage and Network
 - * Moderate entry cost – Easy to scale
- * Data Center in a Box (e.g. HP EcoPod)
 - * Compute Storage and Network in single or few Racks
 - * High entry cost – finite limit to scaling
 - * Start using it almost immediately
 - * Quite mobile
- * Converged Devices (e.g. Nutanix)
 - * Compute Storage and Network in a 2 or 4 rack units
 - * Low entry cost – high scalability
 - * Devices for a mesh
 - * Used by Google

Technologies in a DC: Virtualisation Platforms

- * VMWare
 - * Largest market share in enterprise
 - * Workloads can be:
 - * Seamlessly uplifted from physically layer (Slid from bare metal)
 - * Automatically moved from server to server for load balancing and disaster recovery
 - * Cloned from production to create Dev, Test and DR
 - * Supports all X86 operating systems (e.g., Microsoft, Linux, Novell, Legacy.)

Technologies in a DC: Virtualisation Platforms

- * Hyper V
 - * Growing market share in enterprise
 - * No support for uplifting from physical layer (therefore can not be used to clone production)
- * Amazon EC2
 - * Holy Grail of enterprise despite small market share
 - * No automated support for uplift from physical

Technologies in a DC: Virtualisation Platforms

- * Xen

- * Open source virtual server popular with Linux users in Asia
- * Supports all X86 operating systems (e.g., Microsoft, Linux, Novell, Legacy)

- * Virtual Box

- * Open source owned by Oracle
- * Only complete vertical offering (i.e., Oracle is the only CSP attempting to offer solutions in IaaS, PaaS and SaaS).
- * Supports all X86 operating systems (e.g., Microsoft, Linux, Novell, Legacy)

Open Issues

	Vmware	Hyper V	Amazon EC2	Xen	Virtual Box
VMWare	Full Motion	Convert	Rebuild	Compatible	Compatible
Hyper V	Convert	Compatible	Rebuild	Convert	Convert
Amazon EC2	Rebuild	Rebuild	Compatible	Rebuild	Rebuild
Xen	Compatible	Convert	Rebuild	Compatible	Compatible
Virtual Box	Compatible	Convert	Rebuild	Compatible	Compatible

Considerations of Virtualising Data Center

- * Will the physical site have adequate and appropriate electrical power?
- * Will the physical site have adequate and appropriately concentrated cooling capacity?
- * Will the physical site have appropriate security facilities?
- * Will the physical site have adequate utility backup?
- * Will the consolidated/virtualised platform provide the availability needed for the workloads it will run?
- * Will the consolidated/virtualised platform require new support tools and/or staff skills?

Summary

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 - * Virtualisation for different layers
 - * Technologies in DC